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Using solution X-ray scattering to determine the high-resolution structure and morphology of PEGylated liposomal doxorubicin nanodrugs

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ABSTRACT

Background: Among nanodrugs, PEGylated nanoliposomes loaded with an active agent are of major importance. In this paper we studied the structures and morphology of PEGylated nanoliposomes before and after remote loading with doxorubicin.
Methods: High-resolution structures were obtained by solution X-ray scattering combined with our advanced analysis tools. We studied the PEGylated liposomal doxorubicin (PLD) product Doxil®, and its generics, where remote doxorubicin loading is performed by a gradient of ammonium sulfate, and LC100, a novel PLD under development, where remote loading was done by a gradient of ammonium methanesulfonate. The PLD structures were compared with drug-free nanoliposomes having identical composition.
Results: We determined the membrane electron density profiles of the empty and loaded PLDs, the thickness and density of the PEG layers, and the structure of the drug inside the liposomes.
Conclusions: The liposomal membranes had the same structure for both ammonium salts. We found that the drug formed crystals inside PLDs loaded by ammonium sulfate, whereas it had an amorphous morphology in the PLD loaded by ammonium methanesulfonate. The variations of the drug's structural parameters between the generics of Doxil® are similar to the variations between batches of the same product, suggesting that all these products were structurally similar.
General significance: This paper demonstrates that solution X-ray scattering, when combined with our powerful analysis tools, can determine the high-resolution structure of complex non-crystallized nanoparticle dispersions used in nanomedicine, thereby providing useful physical insights into their functions.

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1. Introduction

Nanodrug formulations have the ability to dramatically modify the kinetics, biodistribution, and release profile of the active pharmaceutical ingredient they contain. Consequently, the selectivity of the drug to its targeted tissue can be improved, its effective dose reduced, and its therapeutic ratio increased (the ratio between the lethal dose in 50% of the subjects to the effective dose in 50% of the subjects). Therefore nanodrugs hold great promise in improving the treatment of many diseases, in particular of cancer and inflammations, in which an enhanced permeability and retention effect (EPR) is observed [1–5]. This effect results from the enhanced accumulation of the nanodrugs in the targeted tissue owing to the mean size of nanodrugs (ca. 100 nm), which is smaller than the mean pore size of the vasculature at the target tissue [1–4,6]. Because of these benefits, many types of nanodrug formulations are currently under development, including liposomes, micelles (proteinic, polymeric, and non-polymeric), emulsions of nano- and submicron sizes, dendrimers, nanotubes of various kinds, and polymeric drug conjugates [7,8]. To increase their circulation time, nanodrugs are often PEGylated, in other words, they are coated by a layer of polyethylene glycol (PEG), of molar mass of 750 g/mol or higher.

Nanodrugs are supramolecular assemblies with various complex structures. This complicates their development and approval for clinical use. The experience gained during the past 25 years of nanodrug development, and extensive evaluations by the regulatory agencies clearly demonstrated the need for understanding the detailed structure, physical and chemical properties, and biological principles associated with these assemblies [5,8]. Therefore, regulatory agencies such as the Food

Abbreviations: EPR, Enhanced permeability and retention; PLD, PEGylated liposomal doxorubicin; FDA, Food and Drug Administration; EMA, European Medicines Agency.

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and Drug Administration (FDA) and the European Medicines Agency (EMA) require an in-depth characterization as a major part of the chemistry, manufacturing and control (CMC), in the development of pharmaceuticals [9]. Among all kinds of FDA-approved injectable drugs that are based on drug delivery vehicles, those based on liposomes are the most developed ones; more than twelve FDA-approved liposomal drugs are in clinical use [10], many of which are nanodrugs [8].

Doxil® is a PEGylated liposomal doxorubicin (PLD), which serves as an anticancer drug for intravenous administration. It is the first FDA-approved nanodrug and the first FDA-approved liposomal drug and has been in extensive clinical use in cancer patients since its worldwide authorization (1996). More than half a million people have been treated with Doxil® so far.

Several points contributed to the success of Doxil®. The presence of the lipopolymer DSPE–PEG in the membrane of the drug carrier provides a highly hydrated, and flexible 2000 mg/mol PEG corona to the nanodrug, prompting prolonged drug circulation time and avoidance of the reticuloendothelial system [11].The high and stable remote loading of doxorubicin by a transmembrane ammonium sulfate gradient allows for selective doxorubicin release at the tumor, with minimal release in plasma and healthy tissues. As the lipid bilayer is in the liquid ordered phase, the drug is retained inside the liposomes during storage and while circulating in the blood, enabling slow and controlled zero-order release kinetics at the tumor. Finally, the mean liposome diameter of 80–90 nm, enables the nanodrug to benefit from the EPR effect. After the liposomes reach the tumor, its microenvironment (which contains ammonia produced by glutaminolysis, which is a unique tumor metabolic pathway) induces tumor selective drug release, improving the therapeutic efficacy [5,12,13].

Since each structural aspect influences the performance of Doxil®, much effort has been devoted to characterize the properties of Doxil® and its "empty" (drug-free) nanocarrier, using different approaches and at different levels of resolution. Two critical aspects for the performance of the liposomes [5,12] were studied in detail: the liposomal membrane and the intraliposome aqueous phase, including the drug's physical state. The bulky and highly hydrated corona of the PEG extending from the lipid bilayer into the aqueous phase is critical for enhancing the liposome steric stabilization, and a major contributor to the EPR effect described above. The hydration of the PEG layer, and hence its conformation, have been estimated using ultrasound measurements [14], zeta potential measurement at different ionic strengths [15], Fourier transform infrared spectroscopy [16], and small angle neutron scattering experiments on empty PEGylated nanoliposomes [17]. These studies showed that according to the density and molecular weight of the coating PEG chains, the polymers can assume either the mushroom or the brush conformations [18]. Lasic et al. [19] investigated the structure of the loaded drug using cryo-transmission electron microscopy (cryo-TEM) and x-ray diffraction, and demonstrated that when doxorubicin is remotely loaded into PEGylated nanoliposomes via a transmembrane ammonium sulfate gradient, long nanocrystals of doxorubicin sulfate in the intraliposome aqueous phase are formed and lead to ellipsoidal liposomes.

Solution small angle x-ray scattering (SAXS) is a powerful tool [20–35], which has been used for investigating the structure of nanodrugs [36], and can provide information on the state of the drug in the nanocarrier. Li and co-workers [37] measured the SAXS from doxorubicin-loaded liposomes containing citrate anions, which were loaded by means of a pH gradient (using a high concentration of sodium citrate at pH 4.0 in the intraliposomal aqueous phase). The authors showed that doxorubicin–citrate complex forms a crystalline phase inside the liposomes. Whereas Li et al. focused on the state of the drug, SAXS experiments can also reveal the very intimate organization of the liposome membrane [35]. In their work on empty nanocarriers, Varga et al. [38,39] developed a scattering model for PEGylated nanoliposomes, based on the model of Brzustowicz and Brunger [40] for vesicles. Their SAXS data analysis showed that the distribution of the polymer between the two leaflets forming the bilayer of the PEGylated nanoliposome is asymmetric. They also discussed the mushroom or brush conformation of the polymer. No drug, however, was encapsulated in the liposomes they studied; hence, only the membrane structure was determined by their analysis.

In this paper, we present high-resolution data analyses of solution SAXS measurements of the PLD dispersions, the first being Doxil® (or Caelyx® — the name for same product sold in Europe), and its generics: Lipodox® and the liposomal research drug DOX-NP™, and, in addition, LC100, a different PLD formulation under development. Both kinds of PLDs share the same lipid composition and size distribution, but differ in the ammonium salt used to drive the remote doxorubicin loading. In LC100 ammonium methanesulfonate replaces the ammonium sulfate used for Doxil® and its generics [41]. LC100 has demonstrated similar therapeutic efficacy in mice and reduced Hand and Foot syndrome in rats [41,42]. To better understand the contribution of the remotely loaded drug to the scattering curves, we also analyzed the scattering data from the corresponding drug-free (or "empty") nanoliposomes with identical lipid composition, intraliposomal salt, and size distribution as in the two kinds of PLDs. In our models, we took into account both the structure of the intraliposomal drug and the structure of the liposome's membrane. We determined the density and thickness of each component of the nanodrug liposomal membrane and the structure and morphology of the encapsulated drug (Table 1). WAXS data revealed the unit cell dimensions of the encapsulated doxorubicin nanocrystal.

We then determined the structural parameters of different batches of the following: Lipodox®, DOX-NP™, and Doxil®/Caelyx®. Our findings showed that the variations between different products (Doxil®/Caelyx®, Lipodox®, and DOX-NP™) are similar to the variations within different batches of a single product (Lipodox® or DOX-NP™). Hence, within our sensitivity and resolution, all the products are structurally similar.

Our study shows that by combining accurate solution X-ray scattering measurements with advanced data analysis tools, detailed structural characterization of nanodrugs such as PLDs can be attained. The findings highlight the importance of the type of ammonium salt anion used for remote drug loading in determining the final structure of the drug, both in the bulk and inside the loaded liposomes, which in turn contributes to the shape of the nanoliposome.

2. Material and methods

2.1. Liposomes

Doxil®/Caelyx® are from Janssen Pharmaceuticals, Inc. (Horsham, Pennsylvania, USA). It is a dispersion of PLD in 10 wt.% sucrose and 10 mM histidine buffer at pH 6.5. The mean size of the PLD is between 75 and 90 nm, and the total lipid concentration is ca. 16.0 mg/ml (21.6 mM). The composition of Doxil®/Caelyx® includes hydrogenated soy phosphatidylcholine (HSPC), cholesterol, and methyl-distearoyl phospho-ethanolamine PEG 2 K (DSPE–PEG) sodium salt in a weight ratio of 3:1:1 (corresponding to mole ratio of 56.51:38.18:5.31), and 2 mg/ml doxorubicin (=3.68 mM), of which more than 90% is liposome-encapsulated. The HSPC is >98% pure and contains ~70% distearoyl PC (DSPC) and ~30% palmitoyl stearoyl PC (PSPC). The solid-ordered to liquid-disordered phase transition of HSPC has its maximum change in heat capacity at its melting temperature, T_m ~52.5 °C [43]. Doxorubicin was remotely loaded to Doxil®/Caelyx® and its generics (Lipodox® and Dox-NP™) using a transmembrane gradient of ammonium sulfate [44,45]. Lipodox® is an FDA-approved generic of Doxil® produced by Sun. Pharma (Mumbai, India) and therefore should be similar to Doxil. Dox-NP™ is a Doxil®-like research reagent (with the same composition, including lipids and drug concentrations, and size distribution) produced by Lipocure Ltd. (Jerusalem, Israel) for Avanti Polar Lipids, Inc., Alabaster, Alabama, USA.

Table 1
Summary of the structural parameters of Doxil®/Caelyx®, Lipodox®, DOX-NP™, and LC100, and the corresponding empty liposomes obtained in this study.

	Empty Doxil®/Caelyx®, Lipodox®, DOX-NP™	Doxil®/Caelyx®, Lipodox®,/DOX-NP™	Empty LC 100	LC 100
Head to head distance (±0.1 nm)	4.80	4.80	4.69	4.69
Inner PEG layer, L_{in} (±1.2 nm)	3.6	3.6	4.1	*
Outer PEG layer, L_{out} (±1.2 nm)	4.8	4.8	4.6	4.3
Drug morphology	_	rod nanocrystal	Amorphous	

* The PEG layer and the inner medium could not be distinguished in LC100 owing to the low electron density contrast between the PEG and the dissolved doxorubicin.

Drug-free PEGylated nanoliposomes, or "empty liposomes", having the same lipid composition and size distribution as Doxil® and its generics were prepared using the protocol of Peleg-Shulman et al. [46]. Lipids (purchased from Lipoid, Ludwigshafen, Germany) were dissolved at high concentration in ethanol. The solution was then mixed with the desired amount of ammonium sulfate aqueous solution (USP grade, Merck) at a temperature above HSPC's T_m of 52.5 °C (preferably above 60 °C). The resulting solution is a dispersion of multilamellar vesicles in 250 mM ammonium sulfate solution at pH 5.5. The multilamellar vesicles were stepwise extruded using a Lipex extruder (Northern Lipids, Inc., Burnaby, BC, Canada) with polycarbonate membranes (GE Water & Process Technologies) using the following sequence of pore diameters: 400, 100, 80, and 50 nm, at a temperature that varied between 60 and 65 °C. After the fourth extrusion, dynamic light scattering measurements of the single unilamellar vesicles gave an average liposome size that varied between 75 and 90 nm and a polydispersity index (PDI) of 0.05 or less.

The transmembrane ammonium sulfate gradient was created by removal of the external ammonium sulfate, replacing it by 4 steps of dialysis at 4 °C against 10% sucrose solution (BioLab Ltd., Jerusalem, Israel), which replaces more than 99% of the external nano-liposomes' ammonium sulfate solution by 10% sucrose solution. The storage temperature was 4 °C and the final storage medium was 10 wt.% sucrose in 10 mM histidine buffer at pH 6.5 (Merck), as in Doxil®/Caelyx®, Lipodox®, or DOX-NP™.

LC100 is a PLD having identical size distribution and composition as Doxil®. Doxorubicin remote loading, however, was driven by ammonium methanesulfonate (rather than ammonium sulfate). "Empty" LC100 liposomes were prepared as described above, replacing the 250 mM ammonium sulfate solution by 350 mM ammonium methanesulfonate solution. The total lipid concentration was 16.0 mg/ml. To produce LC100, the above drug-free liposomes were remotely loaded with doxorubicin HCL (Sicor, Milan, Italy). Remote loading was performed at a temperature above the phase transition temperature of HSPC (preferably above 60 °C) as described [44]. LC100 was stored in 10 wt.% sucrose and 10 mM histidine, pH 6.0, at 4 °C [41].

2.2. X-ray scattering experiments

2.2.1. Small angle X-ray scattering (SAXS)

SAXS measurements were performed in our in-house setup. A detailed description is available in [26]. Because the samples were not oriented, SAXS scans collected on a 2D area detector exhibited a powder pattern. Radial integration of the 2D scattering image was carried out, using FIT2D [47]. The sample-to-detector distance was determined, using silver behenate as a standard, at about 1.860 m. For every measured sample, background measurement of the corresponding reference buffer solution was collected. Both sample and reference buffer background solution were injected into the same flow-through capillary setup, allowing a precise measurement of the background for each sample. During fitting, the instrumental error was taken into account by convoluting the calculated signal with a Gaussian function representing the instrument resolution function, having a full width at half maximum (FWHM) of $\tau_q = 0.08\ nm^{-1}$. All the scattering curves presented in this paper are suitably background-subtracted. Data prior to background subtraction and background measurements are shown in Figure S1.

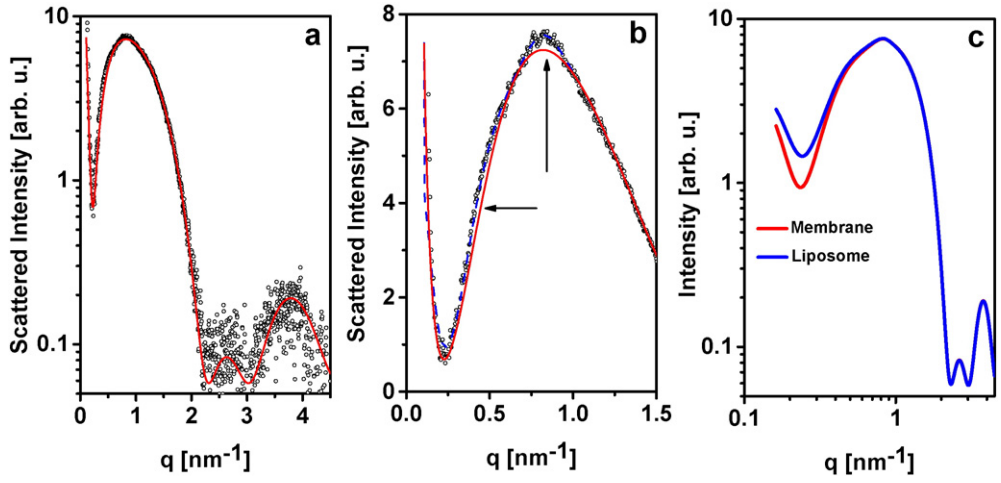


Fig. 1. *X-ray scattering from empty liposome.* a. Background-subtracted radially-integrated solution scattering intensity from a dispersion of 16 mg/ml empty liposomes of Doxil® (black open symbols). The measured signal was fitted to a model of a stack of five infinite flat slabs having Gaussian electron density profiles along the normal to the membrane plane (Eq. (1)). These Gaussian functions correspond to the poly(ethylene-glycol) layers, the hydrophobic lipid tails, and hydrophilic lipid head-groups (red curve). Raw data before background subtraction are given in Fig. S1a. The model parameters that best fit the data are given in Table S1. b: Amelioration of the fitting by the multiplication of the scattered intensity with a structure factor (dashed blue curve) corresponding to a small amount (<1 wt.%) of bi-lamellar liposomes in the dispersion. The peaks indicated by the arrows correspond to a lamellar repeat distance of ca. 15 nm. c. Comparison between the calculated solution scattering intensity of an infinite flat membrane (using Eq. (1)) and a mono-dispersed spherical liposome, 80 nm in diameter, composed of a curved membrane with the same electron density profile as the flat membrane (using Eq. (3)) and the same electron density for the inner and outer media. The main difference between the two models appears in the value of the first minimum, which is very sensitive to the resolution function of the X-ray setup and to the polydispersity in the size of the liposomes. The parameters of the "Membrane model" (red curve), which were used in Fig. 1a, are taken from Table S1. The parameters of the "Liposome model" (blue curve) are given in Table S3.

2.2.2. Modeling solution scattering intensity

The scattering amplitude of nanoparticles in dilute solution, where no interaction is expected between the nanoparticles, is the Fourier transform of their electron density contrast with respect to the solvent. Knowing the electron density contrast provides the exact structure and the size of the nanoparticles. The measured intensity in solution is the square of the scattering amplitude, averaged over all the possible orientations of the particle (assuming the solution is isotropic). We used "X+", software developed in our lab, to evaluate and fit models to our data [20].

To analyze our data we have used the following models:

1. The electron density contrast of the liposome's membrane is modeled by a stack of N (infinite) flat layers that have uniform electron density contrast in the xy plane, but their electron density contrast along the z direction (the direction normal to the membrane) is given by a sum of N Gaussian functions.

The maximum of the electron density contrast, $\Delta\rho_i$, of the ith Gaussian layer, is centered at $z = z_i$, and the FWHM of the ith layer is τ_i. The parameter vector of the ith layer is $\vec{p_i} = (z_i, \tau_i, \Delta\rho_i)$, and the total parameter vector is:

$$\vec{p} = (z_1, \ldots, z_N, \tau_1, \ldots, \tau_N, \Delta\rho_1, \ldots, \Delta\rho_N).$$

The electron density is:

$$\Delta\rho\left(z, \vec{p}\right) = \sum_{i=1}^{N} \Delta\rho_i \exp\left[\frac{-4\ln 2(z-z_i)^2}{\tau_i^2}\right].$$

The intensity in solution is:

$$I\left(q, \vec{p}\right) = \frac{2\pi^3 S}{q^2 \ln 2} \times \sum_{i=1}^{N}\sum_{j=1}^{N} \Delta\rho_j \Delta\rho_i \tau_j \tau_i \cdot \exp\left\{-\frac{q^2\left[\tau_i^2+\tau_j^2\right]}{16\ln 2}\right\} \cdot \cos\left[q(z_i - z_j)\right], \quad (1)$$

where S is a correction prefactor with units of area, and q is the magnitude of the scattering vector. Detailed derivation is given in [33].

2. The total electron density, in cylindrical coordinates, of a cylinder with uniform electron density contrast with respect to the solvent, $\Delta\rho$, length 2 h, and radius, R, is:

$$\Delta\rho\left(\vec{r}, R, h\right) = \Delta\rho \cdot \Theta(|r_\perp| - R) \cdot \Theta(h - |z|)$$

where $r_\perp$ is the projection of $\vec{r}$ on the plane perpendicular to the cylinder axis, and Θ is the Heaviside step function.

The scattering intensity in solution is then given by:

$$I_{cyl}(q, h, R, \Delta\rho) = \frac{4\pi^3}{q^4} \int_0^1 \frac{\sin^2(qhx)}{x^2(1-x^2)} R^2 \Delta\rho^2 J_1^2\left(qR\sqrt{1-x^2}\right) dx \quad (2)$$

where J_1 is the first-order Bessel function of the first kind. Detailed derivation is given in [33].

3. The liposome model is based on a spherical model. The inner medium of the liposome is described by a sphere with uniform electron density contrast $\Delta\rho_{IN}$. The membrane is composed of N multiple spherical shells where each layer has a Gaussian electron density profile along the radial direction r. The center of the ith layer is located at r_i, its FWHM is τ_i, and its electron density contrast at r_i is $\Delta\rho_i$.

Table 2
Averaged radii and full width at half max (FWHM) of the drug radius distribution that best fit the data of PEGylated liposomal doxorubicin from different products and different batches of the same product, obtained by subtracting the membrane contribution from the scattering signal of the loaded PLD. The model used to fit the doxorubicin crystal is a cylinder with a length of 70 nm. The data were fit to the model presented in Eq. (2). Using dynamic light scattering the mean size of the liposomes, (or the Z-averaged size, also known as the cumulants mean) and its polydispersity index (PDI) were obtained. Details on DLS measurements can be found in reference [57].

	Doxorubicin crystal		Liposome	
	R (nm)	FWHM (nm)	Size, Z-average (nm)	PDI
Lipodox JKM 1870	8.4	2.7	77	0.04
Lipodox JKL 6054	8.9	2.2	89	0.09
Lipodox JKG 6055	9.1	2.1	86	0.07
Lipodox JKL 3911	8.7	1.7	85	0.11
Lipodox JKL 6053	8.7	2.2	85	0.08
DOX-NP 037	8.9	2.0	84	0.04
DOX-NP 038	8.6	1.5	80	0.04
DOX-NP 040	8.9	1.8	81	0.06
DOX-NP 0190214	8.6	1.8	76	0.04
Doxil #1,300,667	8.4	2.1	81	0.06
Caelyx #110,943,305	9.1	2.9	88	0.06

The parameter vector of the ith layer is $\vec{p}_i = (r_i, \tau_i, \Delta\rho_i)$, and the total parameter vector is:

$$\vec{p} = (\Delta\rho_{IN}, r_1, \ldots, r_N, \tau_1, \ldots, \tau_N, \Delta\rho_1, \ldots, \Delta\rho_N).$$

The electron density contrast of this model is:

$$\Delta\rho\left(\vec{r}\right) = \Delta\rho_{IN} \cdot \left[\Theta(r) - \Theta(r - R_{eff})\right] + \sum_{i=1}^{N} \Delta\rho_i \cdot \exp\left[-\frac{4\ln 2(r-r_i)^2}{\tau_i^2}\right]$$

where $R_{eff} = r_2 - \frac{\sqrt{\ln(10)/\ln(2)}}{2}\tau_2$, which is defined as the radial position where the electron density contrast of the membrane's second layer ($\Delta\rho_2$), corresponding to the inner headgroup of the lipid bilayer, is at a tenth of its maximum value (see Fig. 9c).

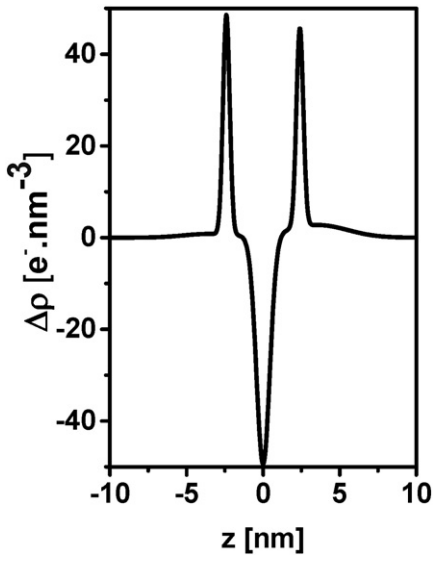


Fig. 2. *Electron density profile of liposome membrane.* Electron density contrast, $\Delta\rho$, of the liposome membrane as a function of z (the direction normal to the bilayer plane) that best fits the data in Fig. 1a (using Eq. (1)). $\Delta\rho$ is modeled with Gaussian functions assuming uniform slabs in the xy plane. $z = 0$ corresponds to the center of the membrane bilayer. Table S1 shows the parameters that best fit the data.

The scattering intensity is:

$$I\left(q,\vec{p}\right)=\left|\begin{array}{l}\frac{4\pi}{q^2}\Delta\rho_{IN}\left[\frac{\sin(qR_{eff})}{q}-R_{eff}\cos(qR_{eff})\right]+\\ \frac{\pi^{3/2}}{2q}\left\{\begin{array}{l}(\rho_1-\rho_{IN})\tau_1 e^{-\frac{\tau_1^2q^2}{16\ln 2}}\left[m_1+\int_0^1\left(g_1\times\exp\left(\frac{\tau_1^2q^2}{16\ln 2}-\frac{4\ln 2r_1^2}{\tau_1^2}\right)y^2\right)dy\right]+\\ \sum_{i=2}^{N}\Delta\rho_i\tau_i e^{-\frac{\tau_i^2q^2}{16\ln 2}}\left[m_i+\int_0^1\left(g_i\times\exp\left(\frac{\tau_i^2q^2}{16\ln 2}-\frac{4\ln 2r_i^2}{\tau_i^2}\right)y^2\right)dy\right]\end{array}\right\}\end{array}\right|^2 \quad (3)$$

where y is an integration variable, m_i is:

$$m_i=-\frac{2r_i\sin(qr_i)}{\sqrt{\ln 2}}-\frac{\tau_i^2q\cos(r_iq)}{4\ln 2\sqrt{\ln 2}}$$

and g_i is:

$$g_i=\left(\frac{4r_i^2}{\tau_i}-\frac{\tau_i^3q^2}{16\ln^2 2}\right)\sin\left[r_iq(y^2-1)\right]-\frac{r_i\tau_iq}{\ln 2}\cos\left[r_iq(y^2-1)\right].$$

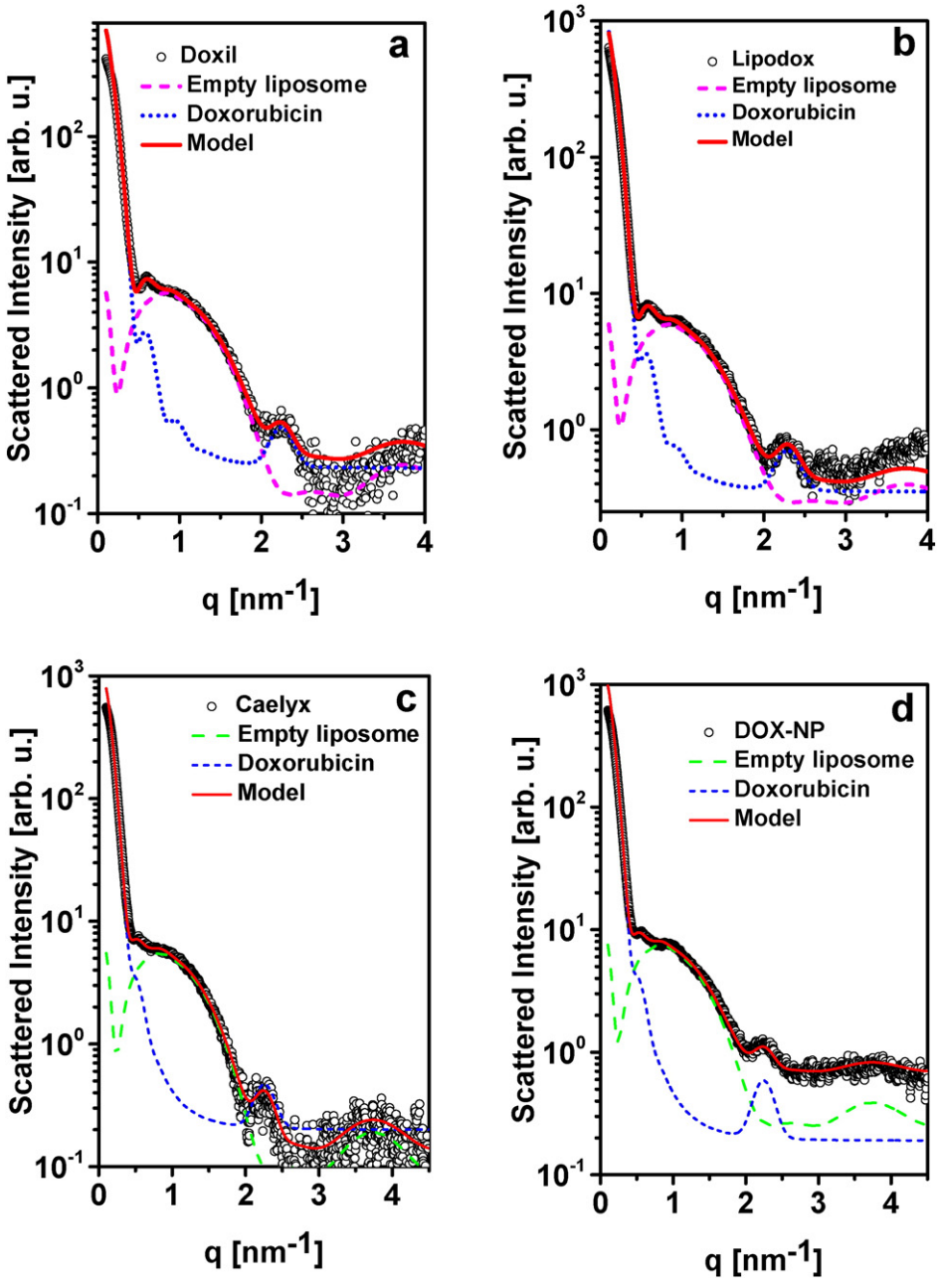


Fig. 3. *Comparison between empty liposomse and liposomes loaded by ammonium sulfate gradient.* Background-subtracted radially-integrated scattering intensity from the nanodrugs Doxil® #1,300,667 (a), Lipodox® #JKL6053 (b), Caelyx® #110,943,305 (c), and DOX-NP™ #037(d) compared with the model that best fits the data of empty liposomes in Fig. 1a. (Broken green curves). The significant overlap of the drug-loaded and unloaded liposome signals indicates that the structures of the membranes are similar with and without doxorubicin. The signal can be decomposed into two parts: membrane identical to the membrane of the empty liposome (dashed green curve) and a 70-nm-long rod-shaped doxorubicin nanocrystal (blue dotted curves). The red curves are the sum of the two combinations (dotted and dashed curves). Data prior to background subtraction and the background solvent signal are shown in Figure S1a. Table 2 presents the parameters that best fit the doxorubicin nanorods in each case.

Detailed derivation can be found in [33].

X+ can also evaluate the effect of polydispersity of each size parameter in each model using a Gaussian distribution function to present the variation in the values of the relevant parameter [20]. The degree of polydispersity is determined by the FWHM of this Gaussian function.

2.2.3. Wide-angle X-ray scattering (WAXS)

Our in-house setup has the flexibility to change the detector position on the beam axis, allowing us to perform wide-angle x-ray scattering (WAXS) experiments. The distance between the sample and the detector is reduced to ca. 0.280 m, and the detector is off-centered. Using this setup, the detector can collect wide-angle x-ray scattered photons corresponding to scattering vectors with a magnitude of up to $q = 35\ nm^{-1}$.

3. Results and discussion

3.1. Remote loading by transmembrane ammonium sulfate gradient

Fig. 1 shows the radially-integrated background-subtracted scattering intensity from a dispersion of empty liposomes with the same membrane composition as Doxil®/Caelyx®, Lipodox®, and DOX-NP™ and the model that best fits the data. Supplementary Figure S2 compares repeated measurements of these liposomes taken from the same batch. Within the noise, the results were essentially identical even though 18 months separated the two measurements. These control measurements demonstrated the stability of our empty liposomes and the reliability of our SAXS measurements. We used X+ to evaluate and fit to our data models of a stack of $N = 5$ (infinite) flat slabs having a Gaussian electron density profile along the direction normal to the membrane of the liposomes (see Materials and Methods, Eq. (1)). Specifically, the two external slabs correspond to the PEG layers inside and outside the liposome; the central slab corresponds to the hydrophobic part of the membrane, whereas the remaining intermediate two slabs, surrounding the central slab, correspond to the hydrophilic lipid headgroups. Each slab is characterized by its maximum electron density contrast, $\Delta\rho_i$, its position relative to the center of the membrane, z_i, and by the FWHM of its Gaussian electron density profile, τ_i. Modeling the electron density profiles of PEG layers with Gaussian function is consistent with experimental results on grafted polymer diffraction experiment [48] and simulations on partly PEGylated lipid surfaces [49]. For the data in Fig. 1a, the model can be ameliorated by introducing a structure factor corresponding to a very small amount (<1 wt.%) of bi-lamellar liposomes in the dispersion (Fig. 1b.). The very weak lamellar peaks correspond to a lamellar repeat distance of ca. 15 nm, which provides an estimate of the thickness of our bilayer with its two PEG layers. We modeled the electron density profile of isolated membrane bilayers rather than liposomes because our fitting results showed that the inner and outer media had electron densities that were very close to one another. Moreover, the mean diameter of the liposomes is about 80 nm (see Table 2), hence the features corresponding to the shape of the liposome could have been detected mainly below our minimal q value, which was ca. 0.1 nm^{-1}. Furthermore, the size of the liposome is somewhat polydispersed, and the size distribution, together with the resolution function of our setup, influence the value of the first minimum in the signal (Fig. 1c). Therefore, our measurement sensitivity and q-range were insufficient to resolve the mean global shape of the liposomes. Fig. 1c demonstrates this point by comparing the expected scattering curves from flat membranes (Eq. (1)) with that of monodispersed spherical liposomes (Eq. (3)) having the same electron density profile.

Fig. 2 shows the profile of the membrane electron density contrast (with respect to the solvent), $\Delta\rho(z)$, which best fits the data in Fig. 1a. $\Delta\rho(z)$ reveals the structural parameters of the liposome. The head to head distance (in other words, the distance from phosphate to phosphate across the bilayer) of the lipid membrane is $z_4 - z_2 = 4.80 \pm 0.10$ nm (based on Table S1). The distribution of lipids in the membrane bilayer is slightly asymmetric. Indeed, an earlier study of small unilamellar vesicles containing cholesterol also showed an asymmetric distribution of the lipids between the two membrane leaflets [50] owing to small curvature differences between them [51]. Moreover, under certain conditions [30], ions can adsorb onto the bilayer surface containing zwitterionic lipid headgroups. In our dispersions, the external medium of the liposome had a low ionic strength (as it contained 10 wt.% sucrose in 10 mM histidine buffer), whereas the inner leaflet faced a high ionic strength ammonium sulfate solution. Ion adsorption could increase the electron density of the inner leaflet and lead to an asymmetric electron density profile. On each side of the liposome the electron density profile of the PEG layer is a bit different. If we define the end of the liposome hydrophilic layer as the position (z), where its electronic density is at 10% of its maximum value ($\Delta\rho_{head}$, in Table S1), and the PEG layer thickness as the distance between the end of the hydrophilic layer and the position where the PEG density is at 10% of its effective maximum

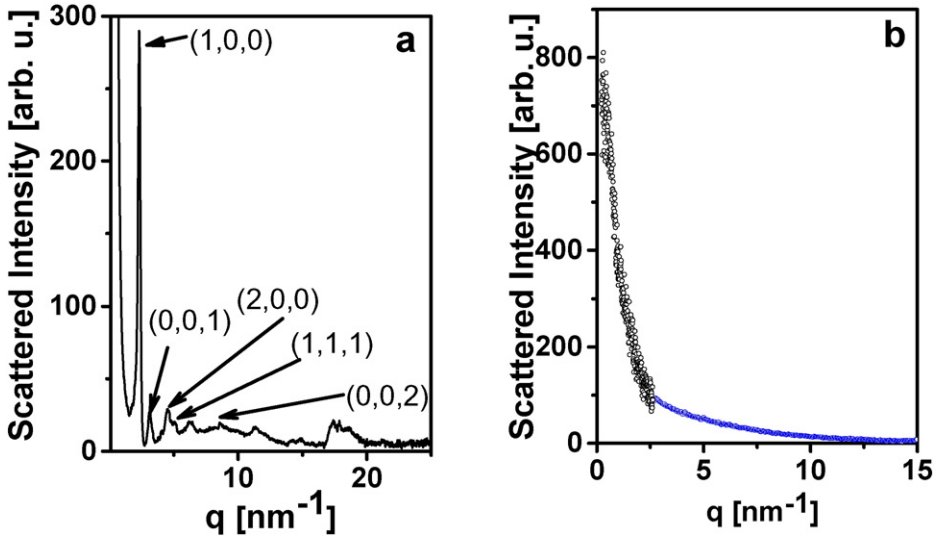


Fig. 4. *X-ray scattering from doxorubicin in ammonium salt solutions.* a. Background-subtracted radially-integrated scattering intensity from a pellet formed in a solution of 25 mg/ml doxorubicin in 125 mM ammonium sulfate. The arrows indicate the scattering peaks that were indexed (in parentheses) to a *3D* hexagonal lattice with unit cell dimensions of $a = b = 3.18$ nm and $c = 2.02$ nm. b. SAXS (black open circles) and WAXS (blue open circles) background-subtracted intensities from a solution of 10 mg/ml doxorubicin in 350 mM ammonium methanesulfonate. No structure-factor peaks were visible.

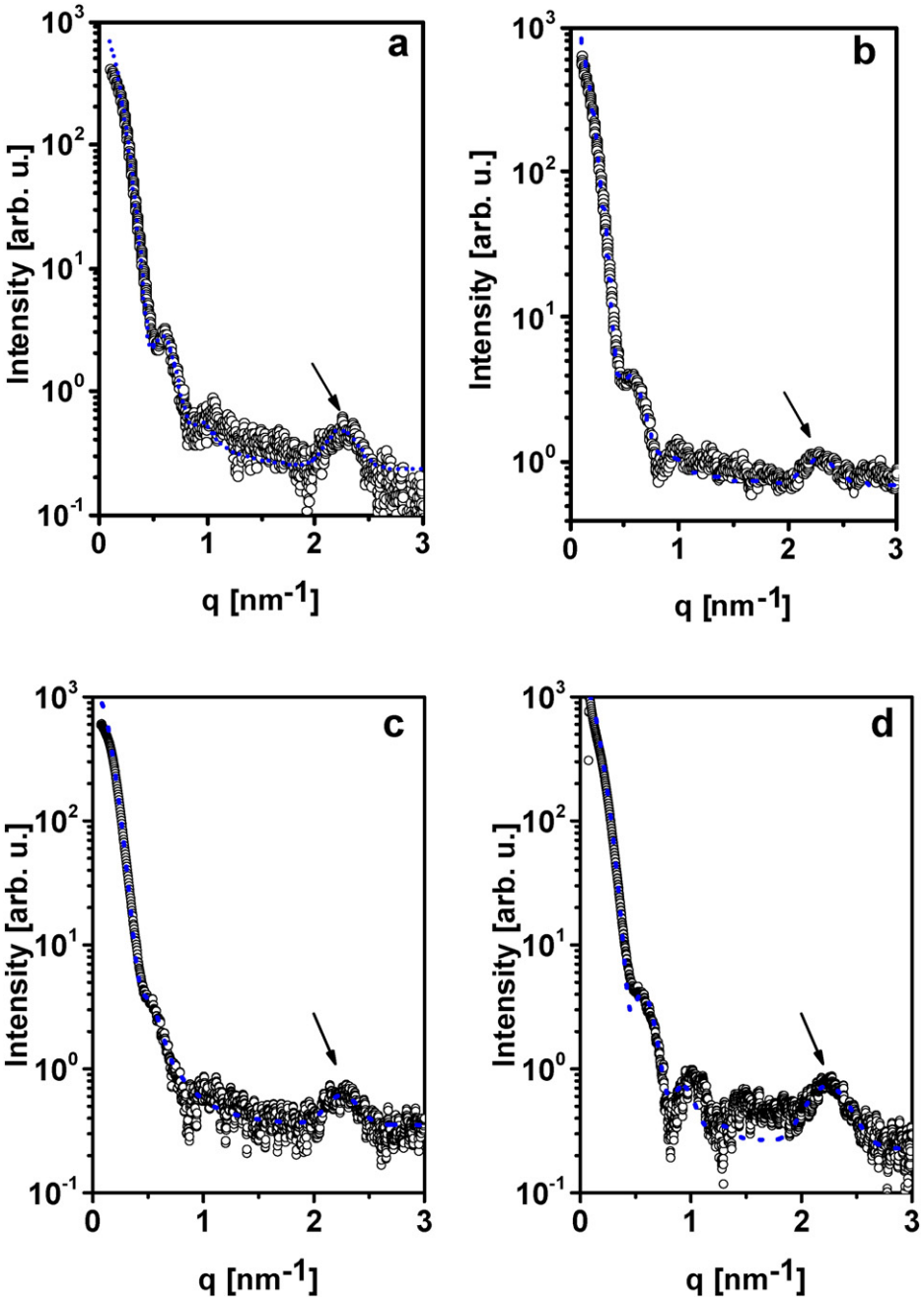


Fig. 5. *X-ray scattering from doxorubicin inside Doxil® and its generics.* The difference between the signals of Doxil® #1,300,667 (a), Lipodox® #JKL6053 (b), Caelyx® #110,943,305 (c), and DOX-NP™ #037(d) and the model that best fit the data of empty liposomes, shown in Fig. 1a (black open symbols). This difference, attributed to the loaded doxorubicin nanocrystal, was fitted to a model of a 70 nm long rod (Eq. (2)), with a polydispersed radius of average value presented in Table 2 (blue dotted curves). The arrows point to the peak in the signals of the doxorubicin nanocrystal at q = 2.25 nm^{-1}, which is almost at the same position as the (1,0,0) peak of doxorubicin dispersion in ammonium sulfate (see Fig. 6a).

value ($\Delta\rho_{PEG}$, in Table S1), then the thicknesses of the PEG layers were 4.8 ± 1.2 nm, and 3.6 ± 1.2 nm for the data in Fig. 1, which were used for the rest of the analysis. The difference between the two PEG thicknesses is consistent with the curvature differences and the asymmetric DSPE–PEG distribution between the two leaflets. When different batches of empty liposomes were measured, similar scattering curves were obtained (Figure S3a and S3b) and, by and large, similar electron density profiles were obtained (Figure S3c and Table S1). The variation in the PEG layer thickness between different batches was 0.4 nm (Table S1) or less. Our model does not differentiate between the inner and the outer layers. It is reasonable, however, to attribute the thicker layer to the outer side of the membrane owing to its lower curvature, hence $L_{out} = 4.8 \pm 1.2$ nm, and $L_{in} = 3.6 \pm 1.2$ nm. The thicknesses of the PEG layers were consistent with recent simulations of PEG chains grafted to lipid membranes [52].

These values should be compared with the Flory radius, R_F, of PEG 2 K in a good solvent, which is the radius of a grafted polymer coil in the mushroom regime, and the height, L, of the polymer brush in the high grafting densities regime in the Alexandre–deGennes theory [53]. The Flory radius is defined by $R_F = aN_m^{3/5}$, where $N_m = 45$ is the number of monomers, a is the size of the monomer unit, and $L = N_m a(a/D)^{2/3}$, where D is the average distance between two grafting points. If we use the value 0.36 nm [54] for a, then $R_F = 3.5$ nm. When $L \sim R_F$, the PEG chains are in the mushroom conformation, hence in the inner layer this is the conformation of the PEG chains, and $D_{in} \geq R_F$ but cannot be estimated using the above relation. Estimating D at the outer layer is

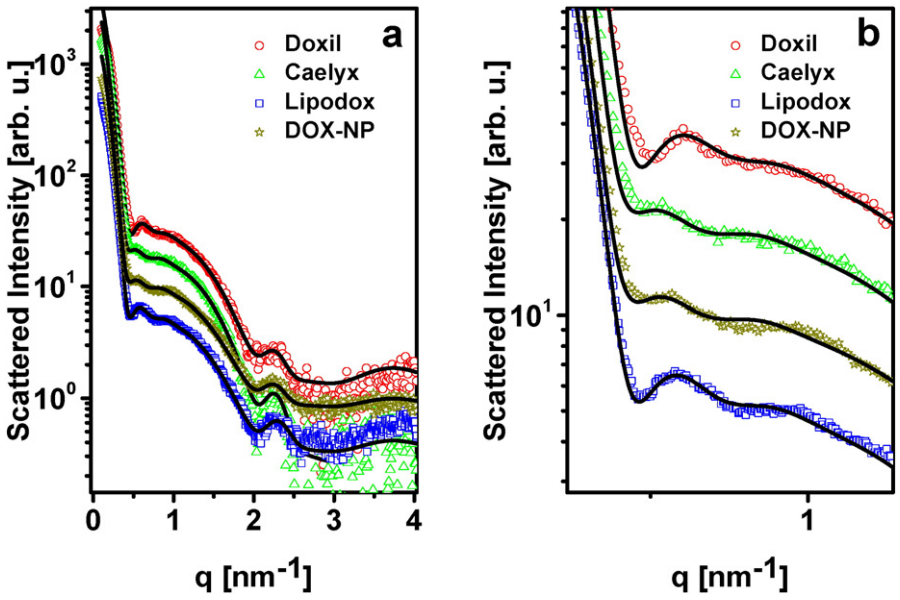


Fig. 6. *Comparing Doxil with its generics.* Comparison between the scattered intensity from Doxil® #1,300,667, Caelyx® #110,943,305, Lipodox® #JKL6053, and DOX-NP™ #037 (open symbols). a. The data and models were multiplied by an arbitrary constant (hence shifted in the logarithmic intensity scale) for clarity of presentation. The solid curves are the best fit of our models (see Fig. 3) to the data. The models are a linear combination of the membrane signal (see Fig. 1a) and a 70-nm-long rod, modeling the doxorubicin nanocrystal with parameters that are presented in Table 2. b: Showing the data at 0.3 $nm^{-1} \le q \le 1.2\ nm^{-1}$ on an expanded scale.

complicated because the liposome membrane contains three components (HSPC, cholesterol, and DSPE–PEG), and the distribution of these components between the inner and outer leaflet and within each leaflet is unknown. Using the experimental mean values of $L_{out} = 4.8$ nm yields $D_{out} = 2.2$ nm, corresponding to a mean area per DSPE–PEG in the outer layer of ca. 5.0 nm^2. The asymmetry in the lipid distribution between the outer and the inner layers and the curvature differences induce an asymmetry of the PEG conformations (mushroom versus brush) at both sides of the liposome membrane. Other preparations of the same empty liposomes have been measured (see Figure S3), and the fitting parameters are within the error bars.

When doxorubicin was loaded by means of transmembrane ammonium ion gradient with sulfate as the ammonium bivalent counteranion, as in Doxil® (Fig. 3a), Lipodox® (Fig. 3b), Caelyx® (Fig. 3c), and DOX-NP™ (Fig. 3d), the scattering intensity of the dispersion increased at low q compared with the data or model that best fits the scattering intensity from the corresponding empty liposomes (Fig. 1). Nevertheless, at $0.8\ nm^{-1} < q < 1.8\ nm^{-1}$ and at $q > 2.8\ nm^{-1}$, the curves of the empty and drug-loaded liposomes overlapped (Fig. 3). For all 4 products, this overlap is a strong indication that in the loaded and unloaded liposomes, the structure of the lipid bilayer and the organization of the lipids in the bilayer were, within our resolution, similar.

To determine the structure of the drug that was remotely loaded into the intra-liposome aqueous phase, we measured the WAXS from 25 mg/ml doxorubicin hydrochloride dispersed in 125 mM ammonium sulfate solution (Fig. 4a). We used higher drug concentrations than the initial bulk-loading concentration (2 mg/ml or ca. 3 mM) because inside the liposomes the drug concentration is ca. 250 mM, owing to the mechanism by which ammonium transmembrane gradient drives doxorubicin remote loading [44,55].

The scattering pattern showed several diffraction peaks, with the first and strongest peak at $q = 2.30\ nm^{-1}$. A similar result was published previously by Lasic et al. [19]. The arrows in Fig. 4a show the expected peak positions of a 3D hexagonal lattice with dimensions $a = b = 3.18$ nm, and $c = 2.02$ nm. Note that in ammonium sulfate solution, the chloride anions of doxorubicin hydrochloride are exchanged for the sulfate anions, which become part of the crystal formed. For the loaded liposomes, (Fig. 3), the (1,0,0) peak of doxorubicin at $q = 2.30\ nm^{-1}$ was clearly visible, indicating that inside the liposomes the doxorubicin was in a crystalline state, and that its lattice was similar to the free doxorubicin–sulfate crystal. The FWHM values of the diffraction peaks, however, were greater than in the bulk (Fig. 4a) owning to the reduced domain size of the doxorubicin crystal in this confined environment. Note that in the preparation the excess free drug was washed out, hence the only contribution to this peak was from the drug inside the liposomes. These results are in agreement with our recent studies on the thermotropic behavior of Doxil®/Caelyx® and Lipodox® in which a narrow endotherm attributed to the melting of the doxorubicin–sulfate nano-crystal, having a T_m of 69–70 °C, was demonstrated by "high-sensitivity" differential scanning calorimetry [56]. This melting was reversible. The intraliposome doxorubicin–sulfate nano-crystals endotherms were much narrower and more symmetric and therefore had significantly larger "cooperative unit" than the melting of the bulk crystals obtained in 250 mM ammonium sulfate [56].

In the case of remote loading driven by trans-membrane ammonium sulfate gradient, we modeled the signal of the loaded liposome as a sum of the scattered intensity from the liposome membrane, which is identical to the empty one (using Eq. (1) and fitting parameters that best fit the data in Fig. 1), and the scattering intensity from the drug that is loaded into the liposome. The assumption in this approximation is that there were no specific interactions between the membrane and the drug that fixed the position of the drug nanorod crystal with respect to the membrane. The quality of the fit retrospectively supported this assumption. The background-subtracted scattering signal from the empty liposome of Fig. 1 has been subtracted from the background-subtracted scattering signal of the loaded liposome (Fig. 5). We attributed the resulting signal to the contribution of the doxorubicin–sulfate nanorod crystal inside the loaded liposome. This signal was fitted to a form factor of a 70-nm-long cylinder (Eq. (2)). We accounted for the radius polydispersity by assuming a Gaussian distribution, using our software X+. The parameters used for the fitting are summarized in Table 2. This polydispersity, however, might have been overestimated (see Figure S4 in Supporting Information section 5), but within the resolution of our data this assumption seemed adequate. Subtracting the empty liposomes from other batches (Figure S3a) did not change the radius of the doxorubicin nanorod (Figure S5).

The subtracted signal presented also the first harmonic of the free doxorubicin–sulfate complexes structure factor (the peak at

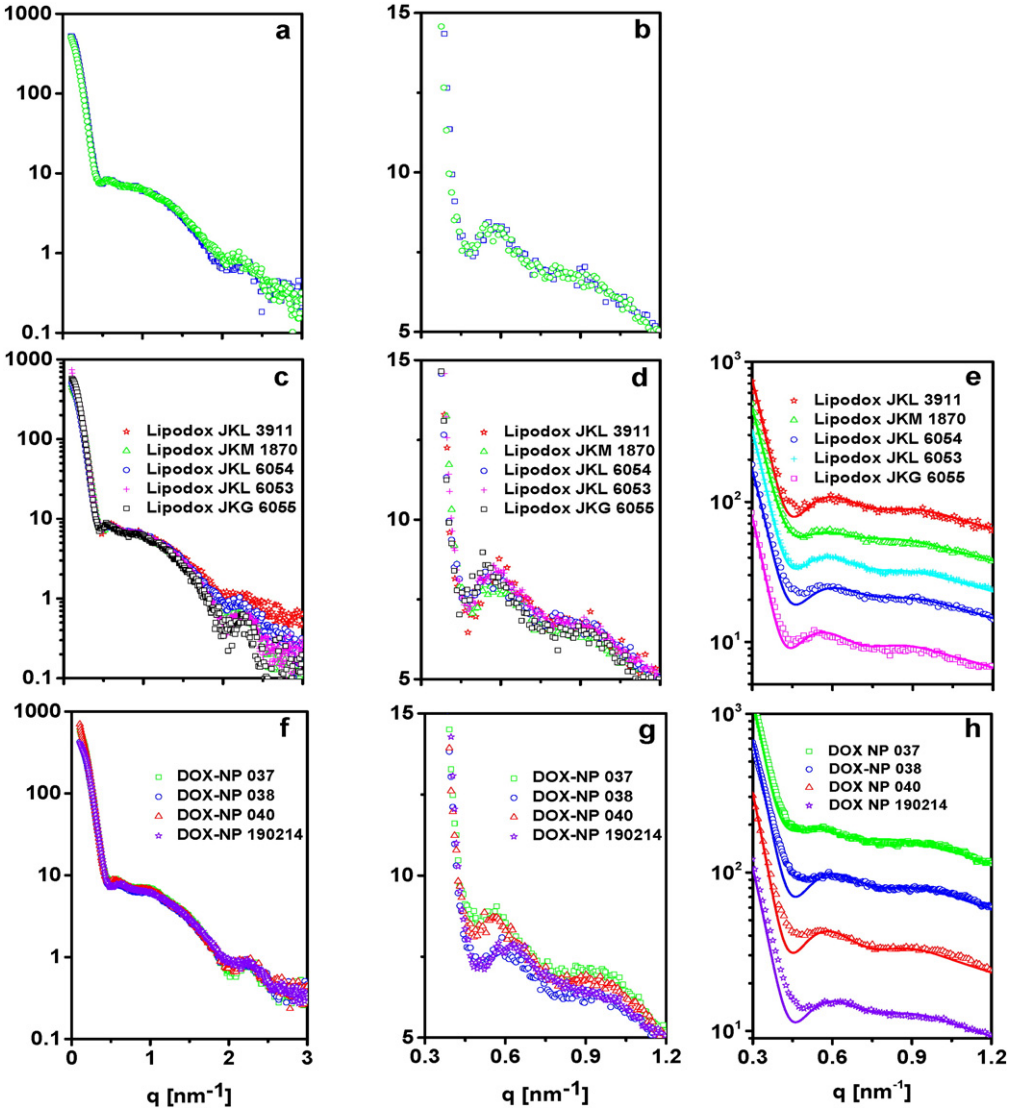


Fig. 7. *Scattered intensity from PLD remotely loaded by ammonium sulfate gradient that are generics of Doxil®.* **a**. Repeated measurements from the same batch of Lipodox® JKL 6054 taken at an 18 month interval. At high q values, we attributed the very small differences between the signals to the slightly different background of the two measurements. **b**. The same data as in a. on an expanded scale. Within the noise, the data are essentially identical. **c**. Comparison between the scattered intensity of different Lipodox® batches. Small differences are visible around the first oscillation. **d**. Same data as in **c** on an expanded scale. **e**. The best fit of those data using the model described in the text. **f**. Comparison between the scattered intensity of different DOX-NP™ batches. **g**. The same data as in **e** on an expanded scale: We attribute the observed differences in the data to the different radii of the doxorubicin cylindrical rods, presented in Table 2. **h**. Best fit of those data using the model described in the text.

q ~2.3 nm^{-1}). Supporting this model, cryo-TEM images [5,57] clearly show that the doxorubicin–sulfate complexes assume an elongated fiber-like structure. In our models, we added the intensities of the membrane and the drug, and not their scattering amplitudes. In other words, we assumed that there were no interferences between the scattering amplitudes of the liposomes and the drug. This assumption was justified by the large aspect ratio of the drug fibers, which ensures that the contact area between the membrane and the drug complex is very small compared with the surface area of the inner leaflet of the liposomal membrane.

In the aforementioned cryo-TEM images, the shape of the liposomes after loading appears ellipsoid (or "coffee bean shape" [19]). The spherical shape of the liposome, however, influences the scattering signal only at low q, (Fig. 1 c). Therefore this change cannot influence the data. Moreover, despite the change in the shape of the liposome, we have seen that the electron density profile of the membrane of the empty liposome is the same as that of the loaded one (Fig. 3).

We then carefully compared Doxil®/Caelyx®, Lipodox®, and DOX-NP™ (Fig. 6). We found in all cases that in a large part of the scattering curves, there is a very good overlap between the scattering data from the empty liposomes (Fig. 1a), and the scattering data from the drug-loaded liposomes (Fig. 3). Hence, in all cases the contribution of the empty bilayer was assumed to be that of Fig. 1a and the differences between the different products were attributed to the structure of the doxorubicin nanorod (Table 2).

We then compared different batches of Lipodox® (Fig. 7c, d, and e) and different batches of DOX-NP™ (Fig. 7f, g, and h), using a similar approach. Table 2 summarizes the radii of the doxorubicin nanorods in each of these products. The results quantitatively show that, within our scatter, the differences between different batches of the same product are similar to the differences between different products, suggesting that, on average, all the products were essentially structurally similar.

Within the same batch, however, our measurements (Fig. 7a and b) showed that the similarity is much higher, demonstrating that the sensitivity of our measurements is very high and the small differences between the signals were owing to real structural differences that averaged out to give similar mean values. We can therefore conclude that within the high sensitivity of our measurements, all the products are

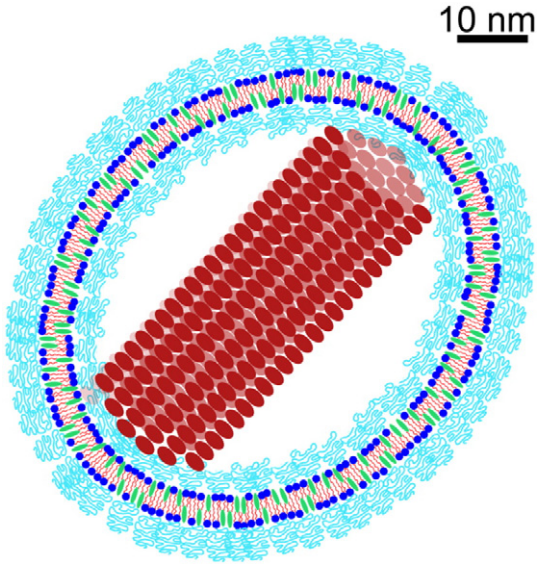


Fig. 8. *Illustration of the structure of Doxil® and its generics as determined by our SAXS and WAXS analyses.* The head-groups of HSPC and the phosphoethanolamine moiety of the DSPE–PEG are shown in blue and their acyl tails are drawn in red, PEG chains are in cyan, cholesterol molecules are in turquoise, and the unit cells of doxorubicin (see Fig. 4) are shown in bourdeaux. The illustration is drawn to scale as indicated by the scale bar and was created by Einav Raviv.

structurally similar. Fig. 8 is drawn to scale and illustrates the structure of Doxil® and its generics as determined by our analyses.

3.2. Remote loading by transmembrane ammonium methanesulfonate gradient

In the liposomal preparation LC100, the inner medium of the liposome contained a solution of 350 mM ammonium methanesulfonate. Unlike the divalent sulfate anion, the methanesulfonate anion is monovalent. The change of counteranion led to drastic differences in the drug structure, as the scattering signals of the two liposomal preparations showed (Fig. 9).

We dispersed 10 mg/ml doxorubicin in 350 mM ammonium methanesulfonate solution and performed SAXS and WAXS measurements (Fig. 4b). The data did not show any scattering peak, suggesting that in the bulk the drug was in an amorphous state under those conditions. We then measured the scattering signals from dispersions of empty liposomes containing only ammonium methanesulfonate solution and from liposomes after loading with doxorubicin by a gradient of ammonium methanesulfonate. With LC100, as with Doxil®, the scattering curves of the loaded and empty liposomes could be superimposed at $q > 1.1\ \mathrm{nm}^{-1}$ (Fig. 9b), indicating that the lipid organization within the membrane is similar in loaded and empty liposomes. The data clearly showed that although the drug was highly concentrated in the LC100 intra-liposome aqueous phase, it did not form any nanocrystal. The presence of the drug, however, influenced the scattered intensity at $q < 1.1\ \mathrm{nm}^{-1}$. This was modeled by changing the electron density of the solution enveloped by the liposome's membrane.

Fig. 9c shows electron density profiles used to fit the measured empty and loaded LC100 liposomes. We used the liposome model (Eq. (3)) with a radius of 40 nm at the center of the lipid bilayer ($r_3 = 40$ nm). Whereas for the empty liposome, the inner medium had the same electron density as the outer medium, for the loaded LC100, the inner part of the liposome had a somewhat higher uniform electron density. After introduction of doxorubicin, the maximum electron density contrast of this part increased by 1.2 e^-/nm^3. Table S2 shows the best fit parameters of these data.

The structures of the loaded and unloaded liposome membranes were almost identical. The head-to-head (in other words, phosphate-to-phosphate) distance across the lipid bilayer was 4.69 ± 0.10 nm, nearly identical to the size obtained for the liposomes loaded by ammonium sulfate gradient, which is not surprising as the signal of both preparations were similar (see Figure S6). For both the loaded and unloaded liposomes, the width of the polymer layer on the outer part of the liposome was similar (4.6 ± 1.0 nm for the empty liposome against 4.3 ± 1.0 nm for the loaded one). As the contrast between the inner PEG layer and the inner medium of the vesicle was very low, it was difficult to estimate the width of the inner PEG layer. We can, however, see from Fig. 9c that, for the empty liposome, the PEG-lipid molecules were distributed asymmetrically between the two liposome lipid leaflets, with more PEG chains in the external layer, which also has a slightly lower curvature. The lack of drug crystals inside LC100 liposomes is consistent with cryo-TEM images [57]. Table 1 summarizes the structural data obtained from this study.

The divalent sulfate anions are able to form salt bridges and support the formation of doxorubicin–sulfate crystals, whereas the monovalent methanesulfonate anions are unable to do so. To understand this, we need to consider the basic unit composed of two opposing doxorubicin–methane cations (DOX–NH_3^+) and their two monovalent methanesulfonate anions. In total, in this unit, there are two attractive interactions between each DOX–NH_3^+ cation and its counterion and two repulsive interactions between the two DOX–NH_3^+ cations and between their two counterions. When, instead of the two monovalent anions, there is a single divalent anion, the latter repulsive interaction between the two monovalent anions does not exist anymore because it has been accounted for by the Born self-energy of the divalent anions. Correlation in the position of the divalent anions results in a net attractive interaction. Although this picture is valid at zero temperature, it most likely holds at the finite temperature of our experiments.

4. Conclusions

We showed that solution X-ray scattering experiments and precise data analysis form a powerful tool for characterizing nanoencapsulated drugs. The electron density profiles of Doxil®/Caelyx®, its generics: Lipodox®, and the research drug DOX-NP™, as well as LC100, and their corresponding drug-free ("empty") PEGylated nanoliposomes have been characterized; the mean thicknesses of the PEG layers and lipid head-to-head (in other words, phosphate-to-phosphate) distances across the bilayer have been determined (Table 1). We clearly showed that the drug hardly affected the structure of the membrane. The lipids in the membrane bilayer were non-uniformly distributed between the two leaflets. In the inner leaflet, the PEG chains assumed mainly mushroom conformations, whereas in the outer layer, the PEG chains assumed more extended conformations of a short polymer brush, or were in the transition region between the mushroom and brush phases. The data were consistent with little or no interaction between the drug and the lipid bilayer.

The extensive control measurements presented here show that solution X-ray scattering is a highly sensitive and noninvasive structural tool that can be used to characterize the high-resolution structure of nanoparticles that are often used in nanobiotechnology and nanomedicine. Once the analysis has been established, this method can readily be used as any other analytical tool for regulatory and/or quality control and assurance.

The structure and morphology of the drug in the bulk were preserved inside both PLDs. We showed that the doxorubicin entrapped in PEGylated liposomes was in a crystalline form if the counterion of the ammomium was the divalent sulfate anion. However, when the monovalent methanesulfonate anion was used as the ammonium counterion, the doxorubicin inside the liposome was in an amorphous or

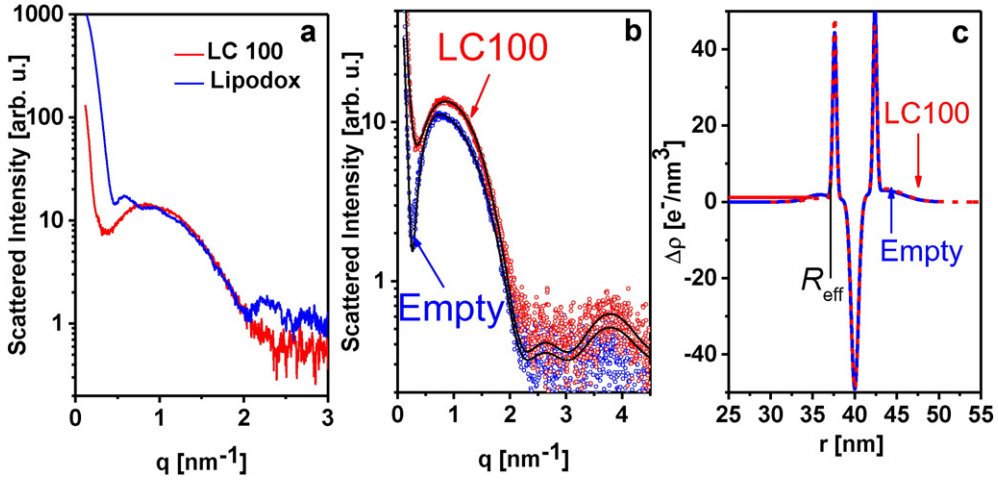


Fig. 9. *X-ray scattering comparisons between LC 100 and Lipodox.* **a**. Comparison between the background subtracted radially integrated scattering intensities of Lipodox® #JKL 6053 (red curve) and LC100 liposomal doxorubicin (blue curve). Unlike the LC100 data, the Lipodox signal exhibited higher intensity and oscillations at low q and a crystalline peak centered at ca. $q = 2.3\ nm^{-1}$. **b**. Background-subtracted radially integrated scattering intensities from LC100 (red circles) and its empty liposome before loading the drug (blue circles). No crystalline peaks were observed, hence the drug that was loaded into the liposomes did not crystallize. The solid black curves show the calculated scattering intensity, using Eq. (3) for LC 100 and Eq. (1) for the liposome membrane before loading. The parameters that best fit the data are shown in Table S2. **c**. The variation of the electron density profiles, along the radial direction (r) normal to the liposome membrane, of the models that best fit the scattering data from the empty liposome (blue) and the loaded one (red). The membrane profile is represented in broken curves, whereas the uniform electron density of inner medium is shown by the solid curves. The radius, r_3, at the center of the liposome in these models was 40 nm. R_{eff} is also indicated by the vertical black line. Table S2 shows the parameters that were used.

liquid state. These findings can be accounted for by the Born self-energy of the divalent sulfate anions, which when they assume positional correlations, contribute to a net attraction between the DOX–NH_3^+ cations and form a salt bridge between them.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.bbagen.2015.09.012.

References

[1] F. Yuan, M. Dellian, D. Fukumura, M. Leunig, D.A. Berk, V.P. Torchilin, R.K. Jain, Vascular-permeability in a human tumor xenograft — molecular-size dependence and cutoff size, Cancer Res. 55 (1995) 3752–3756.
[2] Y. Matsumura, H. Maeda, A new concept for macromolecular therapeutics in cancer-chemotherapy — mechanism of tumoritropic accumulation of proteins and the antitumor agent smancs, Cancer Res. 46 (1986) 6387–6392.
[3] H. Maeda, G.Y. Bharate, J. Daruwalla, Polymeric drugs for efficient tumor-targeted drug delivery based on EPR-effect, Eur. J. Pharm. Biopharm. 71 (2009) 409–419.
[4] R.K. Jain, Transport of molecules in the tumor interstitium — a review, Cancer Res. 47 (1987) 3039–3051.
[5] Y. Barenholz, Doxil® — the first FDA-approved nano-drug: lessons learned, J. Control. Release 160 (2012) 117–134.
[6] H. Maeda, J. Wu, T. Sawa, Y. Matsumura, K. Hori, Tumor vascular permeability and the EPR effect in macromolecular therapeutics: a review, J. Control. Release 65 (2000) 271–284.
[7] D. Peer, J.M. Karp, S. Hong, O.C. Farokhzad, R. Margalit, R. Langer, Nanocarriers as an emerging platform for cancer therapy, Nat. Nanotechnol. 2 (2007) 751–760.
[8] S. Tinkle, S.E. McNeil, S. Muhlebach, R. Bawa, G. Borchard, Y.C. Barenholz, L. Tamarkin, N. Desai, Nanomedicines: addressing the scientific and regulatory gap, Ann. N. Y. Acad. Sci. 1313 (2014) 35–56.
[9] www.fda.gov/downloads/drugs/guidancecomplianceregulatoryinformation/guidances/UCM199635., in.
[10] V.P. Torchilin, Recent advances with liposomes as pharmaceutical carriers, Nat. Rev. Drug Discov. 4 (2005) 145–160.
[11] M.C. Woodle, D.D. Lasic, Sterically stabilized liposomes, Biochim. Biophys. Acta 1113 (1992) 171–199.
[12] Y. Barenholz, Doxil®: the first FDA-approved nanodrug — from an idea to a product (January 2015 update), in: R. Bawa, G. Audette, I. Rubinstein (Eds.), Handbook of Clinical Nanomedicine: Nanoparticles, Imaging, Therapy and Clinical ApplicationsPan Stanford Publishing, Singapore, 2015.
[13] L. Silverman, Y. Barenholz, In vitro experiments showing enhanced release of doxorubicin from Doxil® in the presence of ammonia may explain drug release at tumor site, Nanomedicine (2015) in press.
[14] O. Tirosh, Y. Barenholz, J. Katzhendler, A. Priev, Hydration of polyethylene glycol-grafted liposomes, Biophys. J. 74 (1998) 1371–1379.
[15] K. Shimada, A. Miyagishima, Y. Sadzuka, Y. Nozawa, Y. Mochizuki, H. Ohshima, S. Hirota, Determination of the thickness of the fixed aqueous layer around polyethyleneglycol-coated liposomes, J. Drug Target. 3 (1995) 283–289.
[16] Z. Varga, J. Mihaly, S. Berenyi, A. Bota, Structural characterization of the poly(ethylene glycol) layer of sterically stabilized liposomes by means of FTIR spectroscopy, Eur. Polym. J. 49 (2013) 2415–2421.
[17] L. Arleth, C. Vermehren, An analytical model for the small-angle scattering of polyethylene glycol-modified liposomes, J. Appl. Crystallogr. 43 (2010) 1084–1091.
[18] O. Garbuzenko, Y. Barenholz, A. Priev, Effect of grafted PEG on liposome size and on compressibility and packing of lipid bilayer, Chem. Phys. Lipids 135 (2005) 117–129.
[19] D.D. Lasic, P.M. Frederik, M.C.A. Stuart, Y. Barenholz, T.J. McIntosh, Gelation of liposome interior — a novel method for drug encapsulation, FEBS Lett. 312 (1992) 255–258.
[20] T. Ben-Nun, A. Ginsburg, P. Szekely, U. Raviv, X plus: a comprehensive computationally accelerated structure analysis tool for solution X-ray scattering from supramolecular self-assemblies, J. Appl. Crystallogr. 43 (2010) 1522–1531.
[21] I. Davidi, D. Patra, D. Herrnida-Merino, G. Portale, V.M. Rotello, U. Raviv, R. Shenhar, Hierarchical structures of polystyrene-block-poly(2-vinylpyridine)/palladium-pincer surfactants: effect of weak surfactant-polymer interactions on the morphological behavior, Macromolecules 47 (2014) 5774–5783.
[22] T. Dvir, L. Fink, R. Asor, Y. Schilt, A. Steinar, U. Raviv, Charged membranes under confinement induced by polymer-, salt-, or ionic liquid solutions, Soft Matter 9 (2013) 10640–10649.
[23] T. Dvir, L. Fink, Y. Schilt, U. Raviv, Charging and softening, collapse, and crystallization of dipolar phospholipid membranes by aqueous ionic liquid solutions, Langmuir 30 (2014) 14725–14733.
[24] S. Kler, R. Asor, C. Li, A. Ginsburg, D. Harries, A. Oppenheim, A. Zlotnick, U. Raviv, RNA encapsidation by SV40-derived nanoparticles follows a rapid two-state mechanism, J. Am. Chem. Soc. 134 (2012) 8823–8830.
[25] L. Moshe, G. Saper, O. Szekely, Y. Linde, C. Gilon, D. Harries, U. Raviv, Modulating the structure and interactions of lipid-peptide complexes by varying membrane composition and solution conditions, Soft Matter 9 (2013) 7117–7126.
[26] M. Nadler, A. Steiner, T. Dvir, O. Szekely, P. Szekely, A. Ginsburg, R. Asor, R. Resh, C. Tamburu, M. Peres, U. Raviv, Following the structural changes during zinc-induced

crystallization of charged membranes using time-resolved solution X-ray scattering, Soft Matter 7 (2011) 1512–1523.
[27] G. Saper, S. Kler, R. Asor, A. Oppenheim, U. Raviv, D. Harries, Effect of capsid confinement on the chromatin organization of the SV40 minichromosome, Nucleic Acids Res. 41 (2013) 1569–1580.
[28] A. Steiner, P. Szekely, O. Szekely, T. Dvir, R. Asor, N. Yuval-Naeh, N. Keren, E. Kesselman, D. Danino, R. Resh, A. Ginsburg, V. Guralnik, E. Feldblum, C. Tamburu, M. Peres, U. Raviv, Entropic attraction condenses like-charged interfaces composed of self-assembled molecules, Langmuir 28 (2012) 2604–2613.
[29] O. Szekely, Y. Schilt, A. Steiner, U. Raviv, Regulating the size and stabilization of lipid raft-like domains and using calcium ions as their probe, Langmuir 27 (2011) 14767–14775.
[30] O. Szekely, A. Steiner, P. Szekely, E. Amit, R. Asor, C. Tamburu, U. Raviv, The structure of ions and zwitterionic lipids regulates the charge of dipolar membranes, Langmuir 27 (2011) 7419–7438.
[31] P. Szekely, R. Asor, T. Dvir, O. Szekely, U. Raviv, Effect of temperature on the interactions between dipolar membranes, J. Phys. Chem. B 116 (2012) 3519–3524.
[32] P. Szekely, T. Dvir, R. Asor, R. Resh, A. Steiner, O. Szekely, A. Ginsburg, J. Mosenkis, V. Guralnick, Y. Dan, T. Wolf, C. Tamburu, U. Raviv, Effect of temperature on the structure of charged membranes, J. Phys. Chem. B 115 (2011) 14501–14506.
[33] P. Szekely, A. Ginsburg, T. Ben-Nun, U. Raviv, Solution X-ray scattering form factors of supramolecular self-assembled structures, Langmuir 26 (2010) 13110–13129.
[34] K. Turjeman, Y. Bavli, P. Kizelsztein, Y. Schilt, N. Allon, T.B. Katzir, E. Sasson, U. Raviv, H. Ovadia, Y. Barenholz, Nano-drugs based on nano sterically stabilized liposomes for the treatment of inflammatory neurodegenerative diseases, PLoS One 10 (2015).
[35] D. Zucker, A.V. Andriyanov, A. Steiner, U. Raviv, Y. Barenholz, Characterization of PEGylated nanoliposomes co-remotely loaded with topotecan and vincristine: relating structure and pharmacokinetics to therapeutic efficacy, J. Control. Release 160 (2012) 281–289.
[36] G. Pabst, D. Zweytick, R. Prassl, K. Lohner, Use of X-ray scattering to aid the design and delivery of membrane-active drugs, Eur. Biophys. J. Biophys. Lett. 41 (2012) 915–929.
[37] X.G. Li, D.J. Hirsh, D. Cabral-Lilly, A. Zirkel, S.M. Gruner, A.S. Janoff, W.R. Perkins, Doxorubicin physical state in solution and inside liposomes loaded via a pH gradient, Biochim. Biophys. Acta Biomembr. 1415 (1998) 23–40.
[38] Z. Varga, S. Berenyi, B. Szokol, L. Orfi, G. Keri, I. Petak, A. Hoell, A. Bota, A closer look at the structure of sterically stabilized liposomes: a small-angle X-ray scattering study, J. Phys. Chem. B 114 (2010) 6850–6854.
[39] Z. Varga, A. Wacha, U. Vainio, J. Gummel, A. Bota, Characterization of the PEG layer of sterically stabilized liposomes: a SAXS study, Chem. Phys. Lipids 165 (2012) 387–392.
[40] M.R. Brzustowicz, A.T. Brunger, X-ray scattering from unilamellar lipid vesicles, J. Appl. Crystallogr. 38 (2005) 126–131.
[41] T. Berman, X. Wei, E. Nativ-Roth, S. Peretz-Damari, O. Regev, D. Friedman, Y. Barenhol, Novel pegylated liposomal doxorubicin nano-drug encapsulation using monovalent sulfonic acids, 2015 (in preparation).
[42] Y. Bavli, T. Berman, I. Winkler, Y. Barenholz, A novel pegylated liposomal doxorubicin that reduces Palmar-Plantar Erythrodysesthesia while maintaining the therapeutic efficacy of Doxil/Lipodox against cancer, 2015 (in preparation).
[43] E. Khazanov, A. Priev, J.P. Shillemans, Y. Barenholz, Physicochemical and biological characterization of ceramide-containing liposomes: paving the way to ceramide therapeutic application, Langmuir 24 (2008) 6965–6980.
[44] G. Haran, R. Cohen, L.K. Bar, Y. Barenholz, Transmembrane ammonium–sulfate gradients in liposomes produce efficient and stable entrapment of amphipathic weak bases, Biochim. Biophys. Acta 1151 (1993) 201–215.
[45] Y. Barenholz, Doxil (R) — the first FDA-approved nano-drug: lessons learned, J. Control. Release 160 (2012) 117–134.
[46] T. Peleg-Shulman, D. Gibson, R. Cohen, R. Abra, Y. Barenholz, Characterization of sterically stabilized cisplatin liposomes by nuclear magnetic resonance, Biochim. Biophys. Acta Biomembr. 1510 (2001) 278–291.
[47] A.P. Hammersley, S.O. Svensson, M. Hanfland, A.N. Fitch, D. Hausermann, Two-dimensional detector software: from real detector to idealised image or two-theta scan, High Pressure Res. 14 (1996) 235–248.
[48] G.R. Fleer, M.A. Cohen-Stuart, J.M.H.M. Scheutjens, T. Cosgrove, B. Vincent, Polymers at Interfaces, Chapman & Hall, 1993.
[49] S. Rex, M.J. Zuckermann, M. Lafleur, J.R. Silvius, Experimental and Monte Carlo simulation studies of the thermodynamics of polyethyleneglycol chains grafted to lipid bilayers, Biophys. J. 75 (1998) 2900–2914.
[50] N. Kucerka, M.P. Nieh, J. Katsaras, Asymmetric distribution of cholesterol in unilamellar vesicles of monounsaturated phospholipids, Langmuir 25 (2009) 13522–13527.
[51] D. Lichtenberg, E. Freire, C.F. Schmidt, Y. Barenholz, P.L. Felgner, T.E. Thompson, Effect of surface curvature on stability, thermodynamic behavior, and osmotic activity of dipalmitoylphosphatidylcholine single lamellar vesicles, Biochemistry 20 (1981) 3462–3467.
[52] H. Lee, A.H. de Vries, S.J. Marrink, R.W. Pastor, A coarse-grained model for polyethylene oxide and polyethylene glycol: conformation and hydrodynamics, J. Phys. Chem. B 113 (2009) 13186–13194.
[53] P.G. De Gennes, Polymers at an interface — a simplified view, Adv. Colloid Interf. Sci. 27 (1987) 189–209.
[54] T.L. Kuhl, D.E. Leckband, D.D. Lasic, J.N. Israelachvili, Modulation of interaction forces between bilayers exposing short-chained ethylene-oxide headgroups, Biophys. J. 66 (1994) 1479–1488.
[55] Y. Barenholz, Amphipathic Weak Base Loading into Preformed Liposomes Having a Transmembrane Ammonium Ion Gradient: From the Bench to Approved Doxil, in: Liposome Technology, pp. 1–25.
[56] X. Wei, R. Cohen, E. Nativ-Roth, S. Peretz Damari, O. Regev, Y. Barenholz, Contribution of "high-sensitivity" differential scanning calorimetry (DSC) to elucidating structure/function relationship of the nanodrug, Doxil®, 2015 (submitted for publication).
[57] S. Peretz-Damari, D. Shamrakov, M. Varenik, Y. Barenholz, O. Regev, Nano-liposomes: Size and morphology distributions by combined microscopy and scattering techniques, 2015 (submitted for publication).